Advanced Stats

05-02-26

Population parameters	Sample statistics
size = N	size = n
mean : $\mu = \dfrac{\sum_{i=1}^{N} x_i}{N}$	$\bar{x} = \dfrac{\sum_{i=1}^{n} x_i}{n}$
$\sigma = \sqrt{\dfrac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}}$	$s = \sqrt{\dfrac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}$

if sample is big enough and good enough

big enough → $n > 30$

good enough → good representation of population

has all properties of population

then

$$\bar{x} \approx \mu$$

$$s \approx \sigma$$

Central limit theorem

take m samples, each of size n from any population and compute their means

sample1, sample2, sample3 ..... samplem

means: $\bar{x}_1 \quad \bar{x}_2 \quad \bar{x}_3 \ldots\ldots \bar{x}_m$

The sample means always follow normal distribution

Sample means distribution is normal with

$$\mu_{\bar{x}} = \frac{\sum_{i=1}^{m} \bar{x}_i}{m}$$

$$\sigma_{\bar{x}} = \sqrt{\frac{\sum_{i=1}^{m} (\bar{x}_i - \mu_{\bar{x}})^2}{m}}$$

if the sample size n is big enough then

$$\mu_{\bar{x}} \approx \mu$$

mean of sample means $\approx$ population mean

$$\sigma_{\bar{x}} \approx \frac{\sigma}{\sqrt{n}}$$

std dev of sample means $\approx$ $\frac{\text{population std dev}}{\sqrt{n}}$

n - sample size

Confidence Interval

CI = point estimate $\pm$ margin of error

= mean $\pm$ critical value $\times$ standard error

$$CI = \bar{x} \pm \underset{\substack{\downarrow \\ \text{Z score} \\ \text{table}}}{CV} \times \underset{\substack{\downarrow \\ \frac{s}{\sqrt{n}}}}{SE}$$

How to get critical value / Z score for confidence interval estimation:

1) setup the confidence level

confidence %	confidence level	significance level (α)
95%	0.95	$1 - CL = 1 - 0.95 = 0.05$
98%	0.98	0.02
90%	0.9	0.1

eg. $CL = 0.95$ $\quad SL = \alpha = 1 - 0.95 = 0.05$

sample $\longrightarrow$ mean $= \bar{x}$
std dev $= s$

$\bar{x}$ belongs to a normal dist of sample means

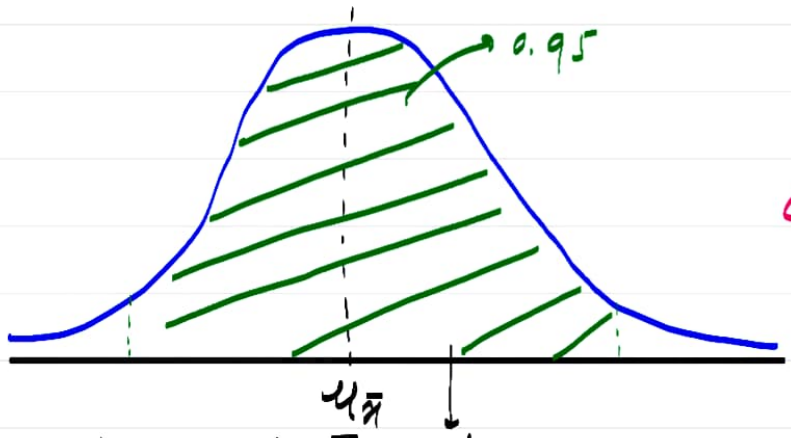
'm' samples $\rightarrow$ means $\underbrace{[\bar{x}_1 \ \bar{x}_2 \ \bar{x}_3 \ \ldots \ \bar{x}_m]}_{\text{follows normal dist.}}$

$\downarrow$

$\mu_{\bar{x}}, \sigma_{\bar{x}}$

$\downarrow$

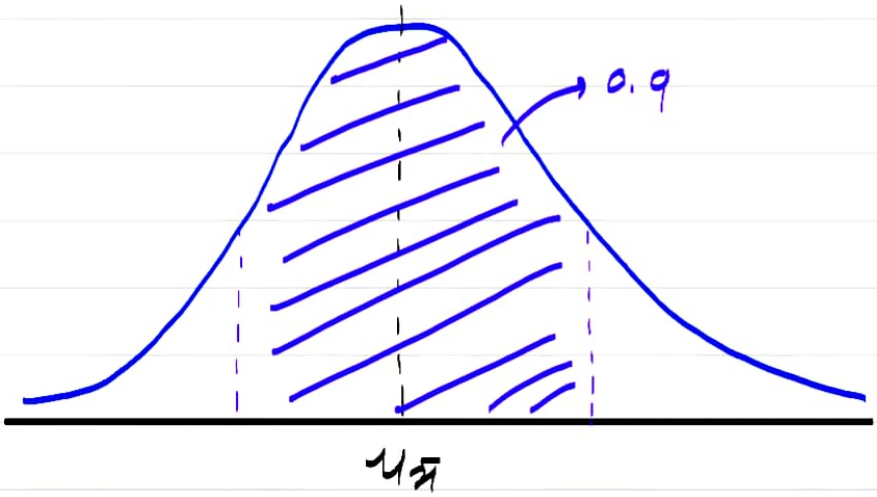
$\mu_{\bar{x}} \approx \mu \rightarrow$ population mean



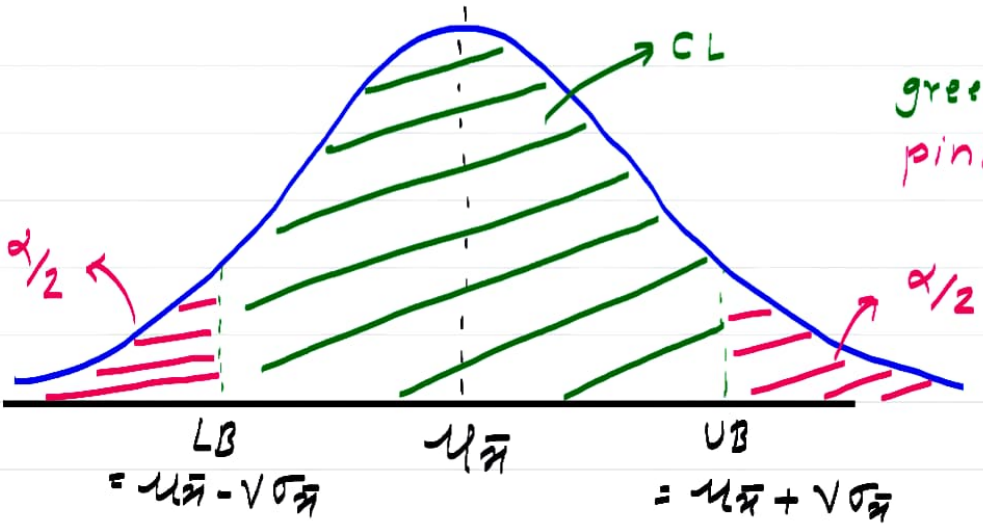
CL = 0.95 is that we want $\bar{x}$ to be present within 0.95 area region around $\mu_{\bar{x}}$

we want $\bar{x}$ to be present in the green area.

if CL is 0.9



we want $\bar{x}$ to be present in purple area for 90% confidence



green area : CL

pink area : $1 - CL = \alpha$

significance level

we want $\bar{x}$ within $\mu_{\bar{x}} \pm V\sigma_{\bar{x}}$

$$\bar{x} \approx \mu_{\bar{x}} \pm V\sigma_{\bar{x}}$$

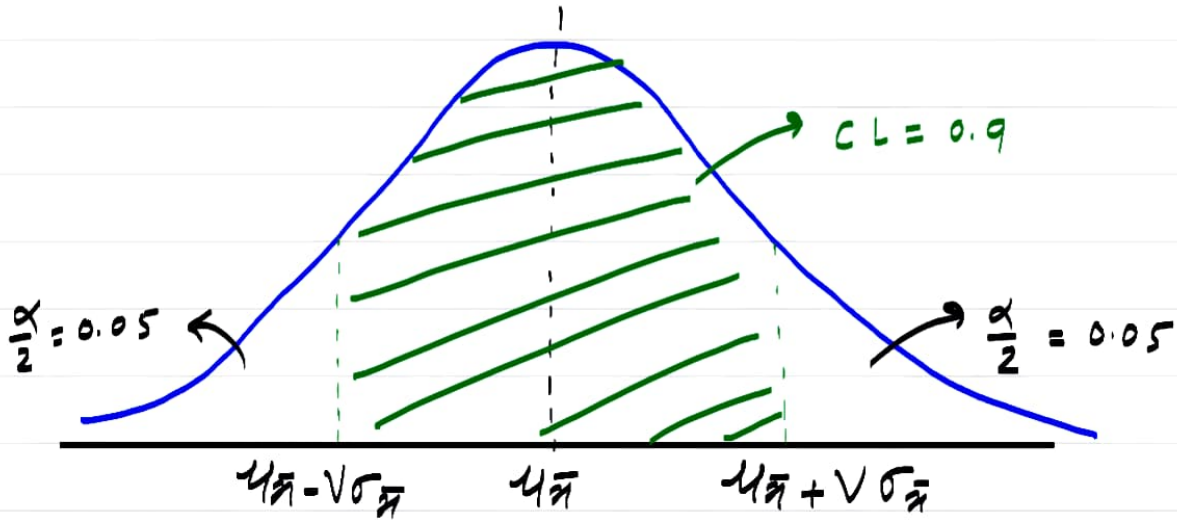
$$\mu_{\bar{x}} \approx \bar{x} \pm V\sigma_{\bar{x}}$$

$$\mu_{\bar{x}} \approx \mu$$

$$\boxed{\mu \approx \bar{x} \pm V\sigma_{\bar{x}}}$$

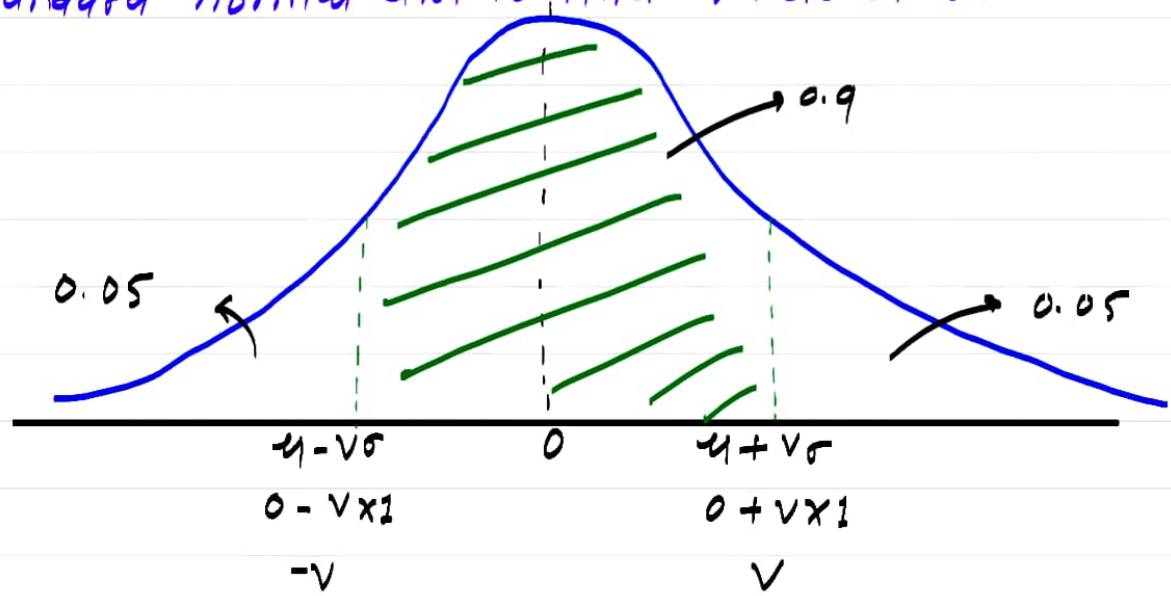
we know $\bar{x}$, $\sigma_{\bar{x}} = \frac{s}{\sqrt{n}}$

we only need V



we need value of V such that green area = 0.9

As area under all normal distributions is consistent, we can use standard normal dist to find value of V.



CDF(V) = area under curve from $-\infty$ to V

CDF(V) = 0.05 + 0.9 = 0.95

we need value of V

CDF(V) = 0.95

$\underbrace{PPF(0.95)}_{\text{Z table gives us the PPF function}} = V$

$CL = C$ sample mean: $\bar{x}$ sample size: n

sample std dev: s

$SL = 1 - C = \alpha$

$$\boxed{CI = \bar{x} \pm PPF\left(1-\frac{\alpha}{2}\right)\frac{s}{\sqrt{n}}}$$

eg. we manufacture iron rods. whats the confidence interval for rods length at confidence level of 95%.

$\longrightarrow$

take a sample of 40 rods and compute sample statistics

$\bar{x} = 2$ m

$s = 0.5$ m

$CL = 0.95$

$SL = 1 - CL = 0.05$

$$CI = \bar{x} \pm PPF\left(1-\frac{\alpha}{2}\right)\frac{s}{\sqrt{n}}$$

$$\boxed{CI = 2 \pm V \times \frac{0.5}{\sqrt{40}}}$$ with $CL = 95\%$

we are 95% confident that avg rod length is betn $\boxed{2 \pm V \times \frac{0.5}{\sqrt{40}}}$